

COMP102: Computer Programming

Practical three

2026

Instructions:

1. Complete all questions.
2. Once completed, get the attention of a tutor during the practical session to mark your work.
3. Students are prohibited from submitting someone else's work as their own for marking.

Question 1

Write a program that:

- Allows the user to specify the number of values (of type double) to be stored in a one-dimensional array.
- Accepts the values from the user and stores them in the array in the order in which they are entered.
- Displays the elements of the array in reverse order on a single line with the values separated by commas.

Question 2

Write a program that:

- Allows the user to enter a message.
- Uses an array to keep track of the number of occurrences of each vowel ('a', 'e', 'i', 'o', 'u') whilst traversing the message.
- Displays the number of occurrences of each vowel after the entire message has been traversed.

Question 3

Write a program that allows the user to create and populate two integer arrays of the same size. The program must:

- Allow the user to specify the size of both arrays.
- Prompt the user to enter the elements of each array in order.
- Clearly indicate which array the user is currently entering values for.

Once both arrays have been populated, perform the following operations in the order given:

1. Calculate the sum of the corresponding elements of the two arrays and store each result at the corresponding index of the **first array**.
2. Calculate the difference between the corresponding elements of the first array and the second array and store each result at the corresponding index of the **second array**.
3. Calculate the difference between the corresponding elements of the first array and the second array and store each result at the corresponding index of the **first array**.
4. Examine the contents of the two arrays after completing the three operations above. What do you notice? **Display both arrays to the console using an appropriate descriptive message.**

Note: *Difference between x and y means $x - y$.*

Question 4

Write a program that allows the user to create and populate an integer array.

The program must:

- Allow the user to specify the size of the array.
- Accept integer values from the user and store them in the array in the order in which they are entered.
- Find the start and end indices of the longest contiguous sequence of equal values for which:
 - The element immediately before the sequence is smaller than the values in the sequence.
 - The element immediately after sequence is also smaller than the values in the sequence.
- Display the start and end indices of the longest sequence found, and the length of the longest sequence.

Note:

- A valid sequence must have an element immediately before it and an element immediately after it.
- A valid sequence must contain at least two elements.
- If more than one sequence has the same highest length, the longest sequence that was found first is given preference.
- If a valid sequence is not found, then display an appropriate message.